Experimental Characterization and Parameters Identification Procedure to generate GaN HeMT Models

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1 Introduction

This document contains the procedure for characterizing a GaN HEMT transistor in the perspective to build a compact SPICE model for the simulation of the transistor switching behavior inside a circuit. It considers that the reader has access to following tools and equipment : curve tracer, SMUs (Source Measure Unit), fixture modules, etc. In particular, the characterization described in this document were carried out with the equipment listed in Tab. 1 and Tab. 2. The Keysight B1404A curve tracer features several SMUs modules, which measure and source voltages and currents simultaneously. The curve tracer also features a CMU (Capacitance Measurement Unit) module, which measures capacitances as a function of DC bias voltage. A summary of the B1505A modules used here is given in Tab. 1.

Module	Mode	Max voltage	Max current	α
MC B1514A	(DC)	±30 V	±100 mA	N/A
MC B1514A	(Pulse)	±30 V	±1 A	$\leq 5\%$
HC B1512A	(DC)	±40 V	±1 A	N/A
HC B1512A	(Pulse)	±20 V	±20 A	$\leq 1\%$
UHC N1265A	(Pulse)	±60 V	±500 A	$\leq 0.1\%$
HV B1513C	(Pulse)	±3 kV	±4 mA	

Table 1 – Voltages and currents limits of B1505AS SMU modules [1].

Curve tracer	HP	HC	MC	UHC	MFCMU
HV 3 kV	✓	✓	✓	✗	✓
UHV 10 kV	✓	✓	✓	✓	✗

Table 2 – Curve tracers configurations.

Module	Frequency range	Voltage range
MFCMU B1520A	1 kHz to 5 MHz	0 V to 25 V
Bias-Tee N1260A	10 kHz to 1 MHz	±3 kV

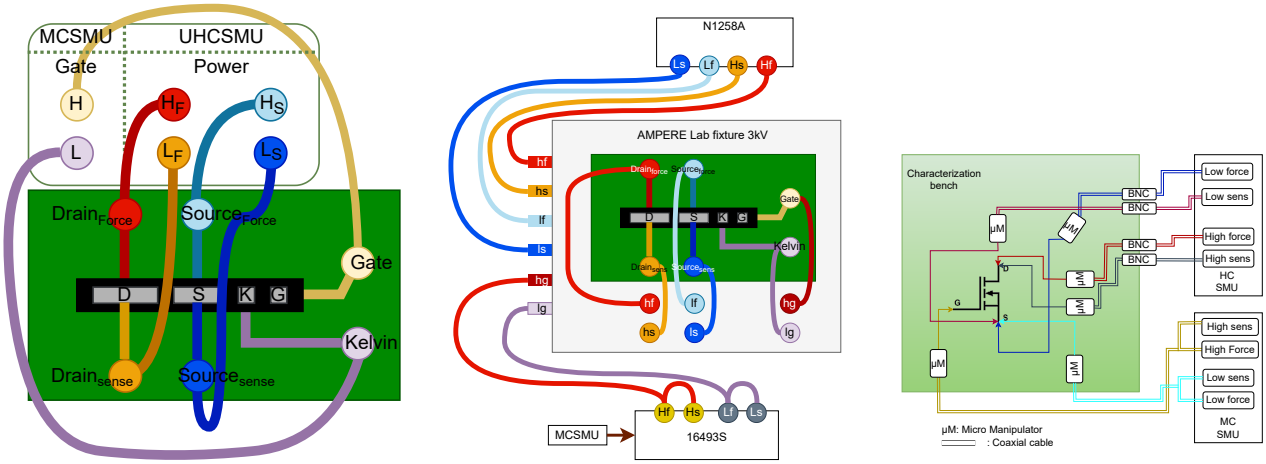
Table 3 – Frequency and voltage limits of the B1505A's CMU [1].

The characterization detailed here comes from two curve tracers (referred to as HV and UHV because of their voltage capability, Tab. 2). Note that both can source voltage levels well in excess of what is required for HeMT characterisation ; both curve tracers also differ in the current levels they can operate at, as well as in the test fixtures they are connected to (Fig. 4).

Before starting any characterization process, several limits must be defined : V_{dsmax}, V_{gsmax}, I_{dmax} (These values are listed in the component data sheet for *commercial devices*, and must be esimated in the case of devices under development).

Curve tracer	Prober	AMPERE Lab Fixture 3 kV	N1258A Module Selector	N1265A UHC Expander
HV 3 kV	✓Fig. 1c	✓	✓	✗
UHV 10 kV	✗	✗	✗	✓ Fig. 1a

TABLE 4 – Components connection configurations



(a) (b) (c)

FIGURE 1 – A four point measurements connection : a - for high current packaged components, b - for low current packaged components, c - for wafer or bare-die components.

2 Selection of device fixture

Based on maximum current value, I_{dmax}, the following curve tracer should be used :
— High current ($I_{d\ max} > 20$ A) → UHV 10 kV
— Low current ($I_{d\ max} \leq 20$ A) → HV 3 kV
Depending on whether the transistors are packaged or not, test fixtures and connections are then chosen as illustrated in Fig. 1.
— High current packaged device - option Fig. 1a
— Low current packaged device - option Fig. 1b
— On-wafer device (or bare die) - option Fig. 1c

3 Device ID and SMU selection

The automated parameter identification procedure which is described further below requires to observe a strict naming convention of the device IDs, Setup Names and variable names, as these are used to retrieve the data files. In this document, we assume data file names are generated by Keysight curve tracers and therefore follow a specific pattern.

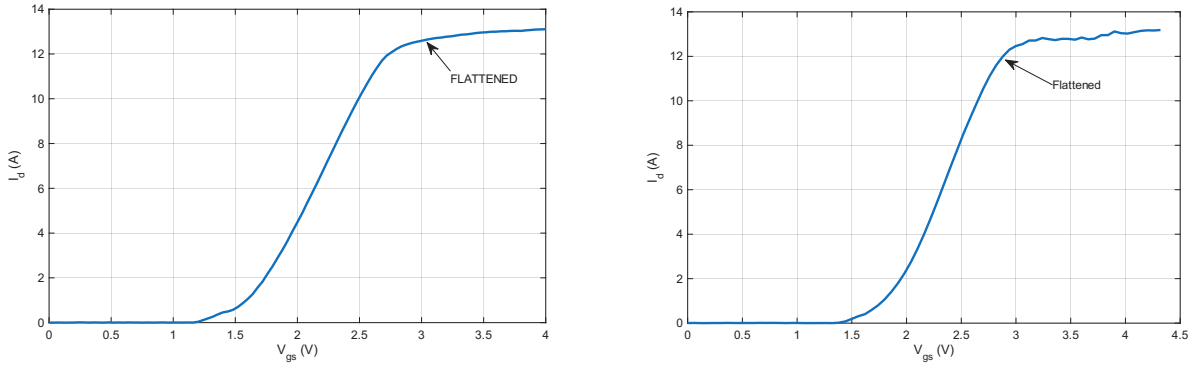
Always use a valid component name by entering a valid and simple *Device ID* for the component to be characterized. example : XYZ.

On the curve tracer B1505A GUI, select the "Multi Channel I/V Sweep". It allows the selection of the Setup Name, SMU, Variables names, modes and functions.

4 Pulsed Id(Vgs)

Variable VAR1 represents the main sweep variable during a measurement.

However, beyond approximately 12 A, the $I_d(V_{gs})$ curves flatten Fig. 2, reducing the drain current's sensitivity and limiting the accuracy of empirical adjustments. Consequently, for drain currents above 12 A, measurements are to be



(a) (b)

FIGURE 2 – a $I_d(V_{gs})$ Curve STMicroelectronics, b $I_d(V_{gs})$ Curve GaNSystem

restricted to $V_{gs_min} \leq V_{gs} \leq V_{gs_lim}$.

V_{gs_lim} is defined such that $I_{dmax}(Fixed\ V_{ds}, V_{gs_lim})$ does not lie in the flattened region.

4.1 Channel Setup Tab

Ensure that the settings in Tab. 5 are set.

Setup Name	Ids-Vgs pulse			
Unit (SMU)	V name	I name	Mode	Function
UHC500	Vds	Ids	VPULSE	CONST
MC	Vgs	Igs	V	VAR1

TABLE 5 – Channels configuration for transfer characteristics (pulsed)

4.2 Measurement Setup Tab

Ensure that the settings in Tab. 6 are applied

VAR1 (primary)					
$Direction = Single$	$No.\ of\ steps = 101$				
Unit :	Name	Start	Stop	Compliance	Pwr comp :
SMU4 :MC	Vgs	V_{gs_min}	V_{gs_lim}	1 mA	OFF
Constants					
Unit :	V Name :	I Name :	mode :	Source	compliance :
SMU2 :UHC500	Vds	Ids	V	$Fixed\ V_{ds}$	$I_{compliance}$

TABLE 6 – Measurement configuration for transfer characteristics pulse-measurement ;

Rq : UHC500 can also be declared as VAR2 Tab. 7.

VAR2 (Secondary)							
Unit :	Name	Start	Stop	Step	#Steps	Compliance	Pwr Comp
SMU3 :UHC	V_{ds}	Fixed V_{ds}	Fixed V_{ds}	1 V	1	$I_{compliance}$	OFF

TABLE 7 – Measurement configuration for drain-source sweep (VAR2)

While $I_{compliance} \geq I_{dmax}$

4.3 Pulse parameters

The pulse parameters can be configured from the *Pulse...* window in the *Measure* tab. To guarantee that the measurements are carried out under consistent circumstances throughout the procedure as listed in Tab. 8.

Common				
Period :	50 ms for UHC500	20 ms for HC		
Measurement Delay Time :	120 µs			
Integration Time				
High Speed ADC :	TIME	20 µs		
Pulse Averaging Count				
Count :	5 to 10			
Pulse Channels				
Unit :	Name	Pulse Delay Time :	Width :	Base :
SMU2 :UHC500	Vds	0 s	150 µs	0 V
or				
SMU3 :HC	Vds	0 s	150 µs	0 V

Table 8 – Pulse parameters

The current measured in the off-state is referred to as I_{noise}

4.4 Display Setup Tab

The data selected in the *List Display* pane will be exported at the end of the measurement. Select Ids, Vds, Igs and Vgs.

5 Id(Vgs) - continuous measurement

Continuous measurements are performed to obtain the transistor characteristics at very low currents (*off-state* several nA), which allows combining them with pulsed characteristics to obtain a full characteristic covering a wide current range.

5.1 Channel Settings Tab

The necessary channels are described in Tab. 9.

Setup Name	Ids-Vds pulse			
Unit (SMU)	V name	I name	Mode	Function
SMU3 :HC	Vds	Ids	V	VAR2
SMU2 :MC	Vgs	Igs	V	VAR1

Table 9 – Channels configuration for transfer characteristic continue

5.2 Measurement Settings Tab

The measurement parameters are described in Tab. 10.
While $10 \times I_{noise} < I_{compliance} < 20 \times I_{noise}$
I_{noise} :The current measured in the off-state in pulsed mode.

VAR1 (primary)					
Direction = Single	*No. of steps = 101*				
Unit :	Name	Start	Stop	Compliance	Pwr comp :
SMU2 :MC	Vgs	V_{gs_min}	V_{gs_lim}	1 mA	OFF
Constants					
Unit :	V Name :	I Name :	mode :	Source	compliance :
SMU3 :HC	Vds	Ids	V	*Fixed* V_{ds}	$I_{compliance}$

TABLE 10 – Measurement configuration for transfer characteristics pulse-measurement ;

5.3 Display Settings Tab

The data selected in the *List Display* pane will be exported at the end of the measurement.
Select Ids, Vds, Igs and Vgs.

6 Id(Vds) - pulse measurement

6.1 Channel Settings Tab

For this measurement configuration, it is mandatory to declare two variables to manage the variation of Vds and Vgs independently as in Tab. 11.

Setup Name	Ids-Vds pulse			
Unit (SMU)	V name	I name	Mode	Function
UHC500	Vds	Ids	VPULSE	VAR1
MC	Vgs	Igs	V	VAR2

TABLE 11 – Channels configuration for output characteristic pulse

6.2 Measurement Settings Tab

The voltage applied to the device under test by the B1505A curve tracer is significantly lower than the voltage at the curve tracer outputs when operating at high current. As a compensation, the drain–source voltage must be swept from $V_{ds} = 0$ up to $V_{ds} = V_{dsmax} + I_{dmax} \times R_{drop}$. We can estimate R_{drop} from the Scope view of the curve tracer Tab. 12.

During the integration time throughout the measurement process, the voltage changes by $\Delta V = 3.6\,\text{V}$ for a drain current of $I_d = 17\,\text{A}$, corresponding to an effective voltage drop resistance of $R_{drop} \approx 0.21\,\Omega$ as illustrated in Fig. 3.

We can therefore conclude that the applied voltage V_{ds_applOC} must be adjusted to account for the internal voltage drop, as given in (1) :

$$V_{ds_applOC} = V_{ds_max} + I_{d_max} \times R_{drop} \tag{1}$$

For this test, we use the same pulse parameters as in Tab. 8
I_{max} is the maximum current the transistor can handle (refer to the manufacturer specifications).

6.3 Display Settings Tab

The data selected in the *List Display* pane will be exported at the end of the measurement.
Select Ids, Vds, Igs and Vgs

7 Id(Vds) - continuous measurement

The purpose of this measurement is to capture the most accurate behavior of the transistor in low-current conditions (off state). Both continuous and pulsed measurements are used in the model parameter identification routine.

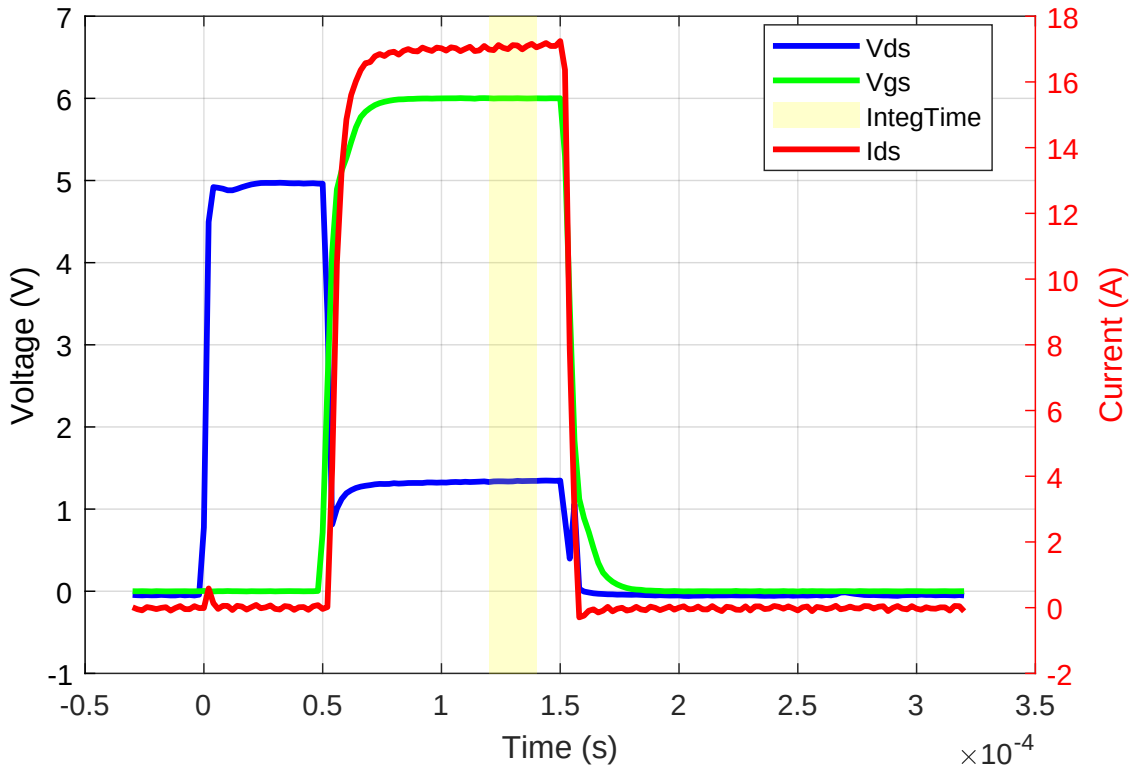


FIGURE 3 – Drain-Source voltage drop

VAR1 (Primary)							
Direction = Single	*No. of steps =*		101				
Unit :	Name	Start	Stop	Compliance	Pwr Comp :		
SMU2 :UHC500	V_{ds}	0 V	V_{ds_applOC}	I_{max}	OFF		
VAR2 (Secondary)							
Unit :	Name	Start	Stop	Step	#Steps	Compliance	Pwr Comp
SMU4 :MC	V_{gs}	V_{gs_minP}	V_{gs_maxP}	1 V	5	1 mA	OFF

TABLE 12 – Measurement configuration for output characteristic pulse

7.1 Channel Settings Tab

The channels are described in Tab. 13.

Setup Name	Ids-Vds pulse			
Unit (SMU)	V name	I name	Mode	Function
SMU3 :HC	Vds	Ids	V	VAR1
SMU2 :MC	Vgs	Igs	V	VAR2

TABLE 13 – Channels configuration for output characteristic pulse

7.2 Measurement Settings Tab

The measurement parameters are described in Tab. 14.

VAR1 (Primary)							
Direction = Single	*No. of steps =*		101				
Unit :	Name	Start	Stop	Compliance	Pwr Comp :		
SMU3 :HC	V_{ds}	0 V	V_{ds_applOC}	10 mA	OFF		
VAR2 (Secondary)							
Unit :	Name	Start	Stop	Step	#Steps	Compliance	Pwr Comp
SMU2 :MC	V_{gs}	V_{gs_minC}	V_{gs_maxC}	1 V	2 or 3	1 mA	OFF

TABLE 14 – Measurement configuration for output characteristic pulse

7.3 Display Settings Tab

The data selected in the *List Display* pane will be exported at the end of the measurement. Select Ids, Vds, Igs and Vgs.

8 Isd(Vsd) - pulsed measurement

8.1 Channel Settings Tab

To measure the reverse characteristics, we keep the same configuration and connections as for the $I_d(V_{ds})$ measurement sec. 6, while adjusting the desired voltage values Tab. 11.

8.2 Measurement Settings Tab

The same principle as the one described in the previous section is applied Tab. 15, taking into account the voltage drop, Rdrop (1) that needs to be compensated for.

VAR1 (Primary)							
Direction = Single	*No. of steps =*		101				
Unit :	Name	Start	Stop	Compliance	Pwr Comp :		
SMU2 :UHC500	V_{ds}	0 V	V_{ds_applRC}	I_{max}	OFF		
VAR2 (Secondary)							
Unit :	Name	Start	Stop	Step	#Steps	Compliance	Pwr Comp
SMU4 :MC	V_{gs}	V_{gs_minR}	V_{gs_maxR}	1 V	4	1 mA	OFF

TABLE 15 – Measurement configuration for output characteristic pulse

I_{max} is the maximum current the transistor can handle (refer to the manufacturer specifications).

8.3 Display Settings Tab

The data selected in the *List Display* pane will be exported at the end of the measurement. Select Ids, Vds, Igs and Vgs.

9 Parasitic capacitance measurements

Measuring internal capacitances is essential, as they directly affect the device's switching behavior and are necessary for accurate modeling.

The measurement circuits are shown in Fig. 4. As mentionned in Tab. 2, the CMU is only availible on 3 kV curve tracer.

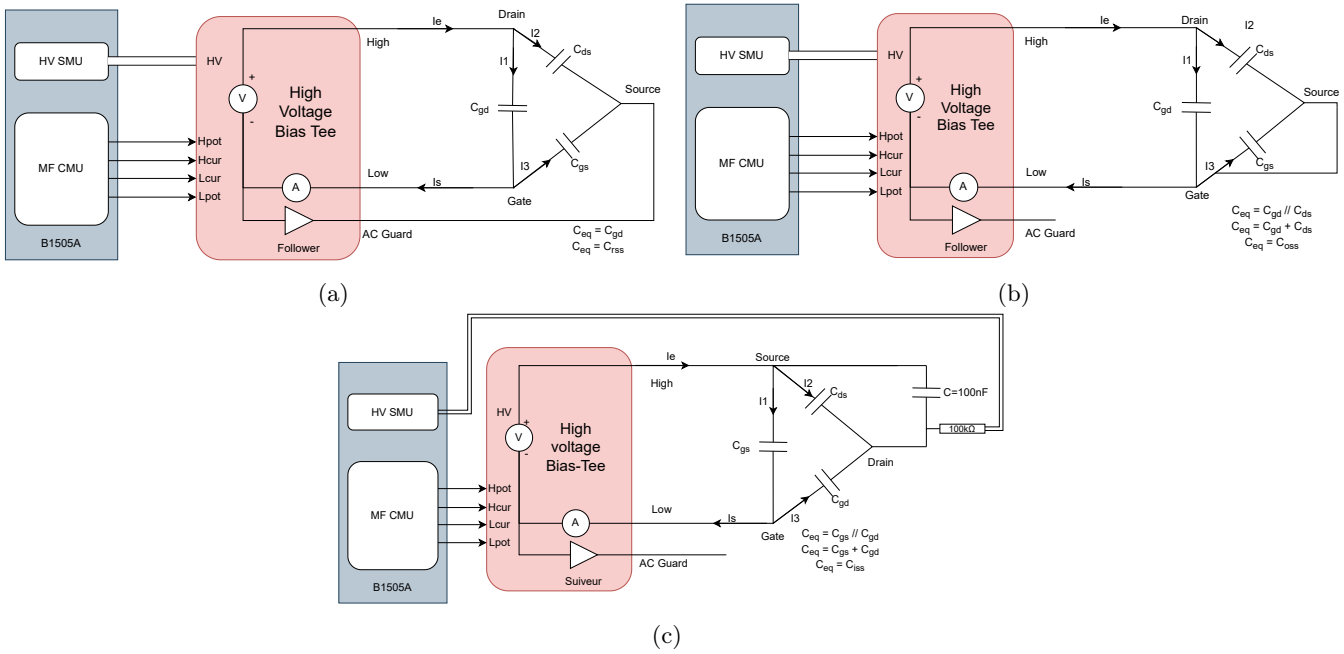


FIGURE 4 – Measurement schematics for the internal capacitances : a - C_{rss}, b - C_{oss}, c - C_{iss}

9.1 Channel Settings Tab

In this case, a Bias Tee Tab. 3 and High Voltage (HV) SMU are used because selected measurement voltages reach up to 650 V for commercial components, and 50 V for on-wafer components Tab. 16.

Setup Name	Crss-Vds	or Coss-Vds	or Ciss-Vds	
Unit (SMU)	V name	I name	Mode	Function
CMU1 :MF	Vd			CONST
HV	Vds	Ids	V	VAR1

TABLE 16 – Channels configuration for Capacitance characterisation

9.2 Measurement Settings Tab

The same configuration is be applied to all measurements, ensuring consistent parameters (number of points, measurement frequencies, etc.) across all tests. The only difference is that the setup names Tab. 16 vary depending on the capacitance being measured Tab. 17.

By default, measurements are performed at 100 kHz **[2]** [1]**. However, other frequency values can be used, for example to match the frequency mentioned by a device manufacturer in their datasheet.**

9.3 Display Settings Tab

The data selected in the *List Display* pane will be exported at the end of the measurement.
Select Vds, C.

10 File name export

It is necessary to ensure that data export is properly enabled for all tests. In the lower-right corner, under 'Auto Export ON/OFF', open the dialog window and check in Data FileTab. 18 :

The curve tracer exports measurement files using a specific naming format Tab. 19. Avoid having multiple files for the same test ; verify that only one file per test exists in the destination folder.

1. Tests confirm that at 100 kHz the results fall within the tolerance range of calibration capacitors, ranging from 1 pF to 1 nF.

C-V (VAR1)		Signal Source	
V Name :	Vds	**Frequency Lis :**	$f_{measurement}$
Model :	Cp-G	**F Name :**	Freq
C Name :	C	**AC Level :**	30 mV
G Name :	G	**Integration Time**	
Direction :	Single	**Mode :**	PLC
Linear/Log	Linear	**Factor**	10
Start :	0V	**Timing**	
Stop :	V_{ds_max}	**Hold :**	10 ms
Step :	$\frac{V_{ds_max}}{100}$	**Delay**	10 ms
Compliance :	8 mA		
No. of steps :	101		

TABLE 17 – Capacitance measurement configuration

✓	Enable automatic data export to the specified storrage device (Auto export ON)
Folder Path :	browse the folder to select the desired file.
File Name	
✓	Default Format
File Type	
✓	CSV
Data Storage	
✓	Enable Automatic data record to the internal storage device (Auto Record ON)

TABLE 18 – Data export parameters

Example for a direct characteristic Ids-Vds in pulse measurement :
$Ids - Vds\ pulse\ [GS66508B(3)\ ;\ 9_1_2025\ 3_55_37\ PM].csv$

Name	Example
Setup Name	Ids-Vds pulse
Device ID	GS66508B
Test number	(3)
Date	9_1_2025
Time	3 :55 :37 PM
File format	.csv

TABLE 19 – Nomenclature of exported curve tracer files

11 Matlab script for compact model building

— All measurement files must be placed in the same folder named after the component **Device ID**
— Once all data are in the same folder, run the MATLAB script `mainScript.m`.
Dedicated instructions can be found in `README_HEMT_Modeling.txt` file.

Références

[1] Hiroyuki Aoyama. “B1505A Power Device Analyzer/Curve Tracer”. In : ().

[2] Salim Dahmani et al. “Characterization and modeling protocol for GaN-on-Si power transistors”. In : *2024 IEEE 11th Workshop on Wide Bandgap Power Devices & Applications (WiPDA)*. 2024 IEEE 11th Workshop on Wide Bandgap Power Devices & Applications (WiPDA). ISSN : 2687-8577. Nov. 2024, p. 1-6. doi : 10.1109/WiPDA62103.2024.10773354. url : https://ieeexplore.ieee.org/document/10773354/ (visité le 24/07/2025).